Individuelle Praktische Arbeit

Sven Toye

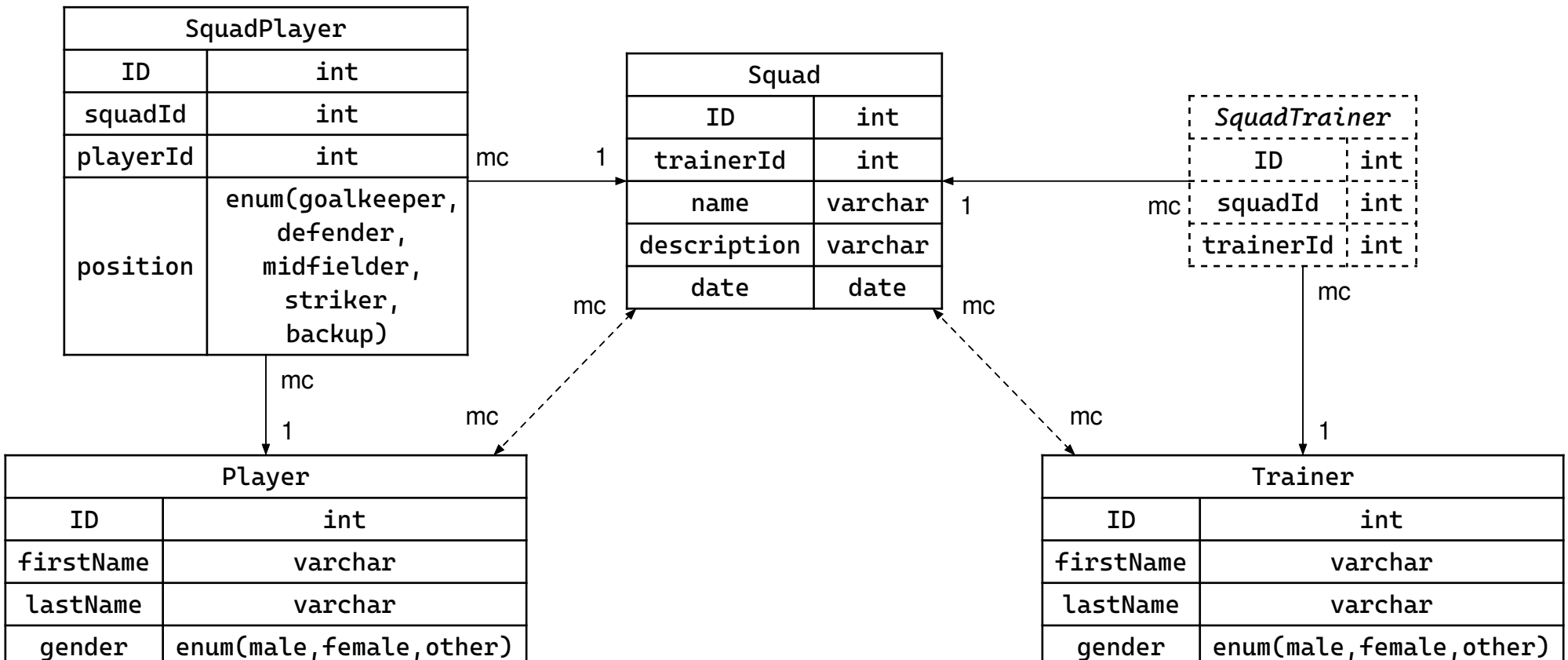
Ausgangssituation

- Eine REST API für die Verwaltung von Spierler*innen, Trainer*innen und Teamaufstellungen.
- Der Fussballverein des VfB Zürich-Leutschenbach verschickt zu jedem Spieltag seiner Jugendmannschaften eine Email an die Spieler/-innen und deren Eltern, in der der jeweilige Mannschaftskader bekannt gegeben wird.
- Um diese Nachricht attraktiver zu gestalten und damit potentielle neue Spieler/-innen zu gewinnen, möchte der Vereinspräsident seinen Trainer/-innen eine WebApp zur Verfügung stellen, mit deren Hilfe sie den Kader des jeweiligen Spieles zusammenstellen können. Dieser Kader kann dann über eine definierte URL aufgerufen und per Email oder Whatsapp verschickt werden.
- Spieler*innen und Trainer*innen können jeweils in mehreren Teamkadern gleichzeitig eingesetzt werden.
- Teamkader müssen nicht aus Spieler*innen des gleichen Geschlechts bestehen, sondern dürfen auch gemischt sein.
- Speiler*innen können verschiedene Positionen in unterschiedlichen haben.
- Folgende Spielpositionen gibt es: Tor, Abwehr, Mittelfeld, Sturm, Ersatzspieler
- Pro Teamkader können mehrere Trainer*innen gesetzt werden.
- Aufgabe dieser PIPA ist es, eine dokumentierte REST Schnittstelle zur Verfügung zu stellen, die beim Bau der Frontend Applikation (NICHT TEIL DIESER API!) verwendet werden kann.

Anforderungen

- Spieler:
  - id
  - first_name
  - last_name
  - gender (male, female, diverse)
- Trainer:
  - id
  - first_name
  - last_name
  - gender (male, female, diverse)
- Squad:
  - id
  - description
  - date
  - player (list of players + positions)
  - trainer (list of trainers)

ERD


SquadPlayer
ID int
squadId int
playerId int
position enum(goalkeeper, defender, midfielder, striker, backup)
Squad
ID int
trainerId int
name varchar
description varchar
date date
SquadTrainer
ID int
squadId int
trainerId int
Player
ID int
firstName varchar
lastName varchar
gender enum(male,female,other)
Trainer
ID int
firstName varchar
lastName varchar
gender enum(male,female,other)
mc
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mc

Player

```ts
@SerializeOptions({ excludeExtraneousValues: true })
@Entity()
class Player {
  @ApiProperty({ description: "ID", example: 1 })
  @PrimaryGeneratedColumn()
  id: number;

  @ApiProperty({ description: "Vorname", example: "Sven" })
  @Column({ length: 255 })
  firstName: string;

  @ApiProperty({ description: "Nachname", example: "Toye" })
  @Column({ length: 255 })
  lastName: string;

  @ApiProperty({ description: "Geschlecht" })
  @Column({ type: "enum", enum: Gender })
  gender: Gender;
}
```

```ts
enum Gender {
  Male = "Male",
  Female = "Female",
  Other = "Other",
}
```

Trainer

```ts
@SerializeOptions({ excludeExtraneousValues: true })
@Entity()
class Trainer {
  @ApiProperty({ description: "ID", example: 1 })
  @PrimaryGeneratedColumn()
  id: number;

  @ApiProperty({ description: "Vorname", example: "Sven" })
  @Column({ length: 255 })
  firstName: string;

  @ApiProperty({ description: "Nachname", example: "Toye" })
  @Column({ length: 255 })
  lastName: string;

  @ApiProperty({ description: "Geschlecht" })
  @Column({ type: "enum", enum: Gender })
  gender: Gender;
}
```

```ts
enum Gender {
  Male = "Male",
  Female = "Female",
  Other = "Other",
}
```

Squad

```
@SerializeOptions({ excludeExtraneousValues: true })
@Entity()
class Squad {
  @ApiProperty({ description: "ID", example: 1 })
  @PrimaryGeneratedColumn()
  id: number;

  @ApiProperty({ description: "Name", example: "FC Zürich" })
  @Column({ length: 255 })
  name: string;

  @ApiProperty({ description: "Beschreibung", example: "Seit 1896." })
  @Column({ length: 255 })
  description: string;

  @ApiProperty({ description: "Erstellungsdatum" })
  @Column()
  date: Date;
  ...
```

```
  ...
  @ApiProperty({ description: "Trainer für diesen Squad" })
  @ManyToMany(() => Trainer, {
    eager: true,
    cascade: true,
    nullable: false,
  })
  @JoinTable()
  trainers: Trainer[];

  @ApiProperty({ description: "Spieler mit ihren Positionen in diesem Team" })
  @OneToMany(() => SquadPlayer, (squadPlayer) => squadPlayer.squad, {
    eager: true,
    cascade: true,
  })
  squadPlayers: SquadPlayer[];
}
```

SquadPlayer

```ts
@Entity()
class SquadPlayer {
  @ApiProperty({ description: "ID", example: 1 })
  @PrimaryGeneratedColumn()
  id: number;

  @ApiProperty({ description: "Position vom Spieler" })
  @Column({ type: "enum", enum: Position })
  position: Position;
  ...
```

```ts
enum Position {
  Goalkeeper = "Goalkeeper",
  Defender = "Defender",
  Midfielder = "Midfielder",
  Striker = "Striker",
  Backup = "Backup",
}
```

```
  ...
  @ApiProperty({ description: "Spieler" })
  @ManyToOne(() => Player, {
    eager: true,
    nullable: false,
    orphanedRowAction: "delete",
    onDelete: "CASCADE",
    onUpdate: "CASCADE",
  })
  player: Player;

  @ApiHideProperty()
  @Exclude()
  @ManyToOne(() => Squad, (squad) => squad.squadPlayers, {
    nullable: false,
    orphanedRowAction: "delete",
    onDelete: "CASCADE",
    onUpdate: "CASCADE",
  })
  squad: Squad;
}
```

Player DTO

```
@SerializeOptions({ excludeExtraneousValues: true })
class CreatePlayerDto {
  @ApiProperty({ description: "Vorname", example: "Sven" })
  @IsString()
  @IsNotEmpty()
  firstName: string;

  @ApiProperty({ description: "Nachname", example: "Toye" })
  @IsNotEmpty()
  @IsString()
  lastName: string;

  @ApiProperty({ description: "Geschlecht" })
  @IsEnum(Gender)
  gender: Gender;
}

class UpdatePlayerDto extends PartialType(CreatePlayerDto) {}
```

Trainer DTO

```
@SerializeOptions({ excludeExtraneousValues: true })
class CreateTrainerDto {
  @ApiProperty({ description: "Vorname", example: "Sven" })
  @IsString()
  @IsNotEmpty()
  firstName: string;

  @ApiProperty({ description: "Nachname", example: "Toye" })
  @IsString()
  @IsNotEmpty()
  lastName: string;

  @ApiProperty({ description: "Geschlecht" })
  @IsEnum(Gender)
  gender: Gender;
}

class UpdateTrainerDto extends PartialType(CreateTrainerDto) {}
```

Squad DTO

```
@SerializeOptions({ excludeExtraneousValues: true })
class CreateSquadDto {
  @ApiProperty({ description: "Name", example: "FC Zürich" })
  @IsString()
  @IsNotEmpty()
  name: string;

  @ApiProperty({ description: "Beschreibung", example: "Seit 1896." })
  @IsString()
  description: string;

  @ApiProperty({ description: "Erstellungsdatum" })
  @IsDate()
  @Type(() => Date)
  date: Date;
  ...
```

```
  ...
  @ApiProperty({ description: "Trainer für diesen Squad" })
  @ValidateNested()
  @IsInstance(TrainerId, { each: true })
  @Type(() => TrainerId)
  trainers: TrainerId[];

  @ApiProperty({ description: "Spieler mit ihren Positionen in diesem Team" })
  @ValidateNested()
  @IsInstance(SquadSquadPlayerDto, { each: true })
  @Type(() => SquadSquadPlayerDto)
  squadPlayers: SquadSquadPlayerDto[];
}

@SerializeOptions({ excludeExtraneousValues: true })
class TrainerId {
  @ApiProperty({ description: "ID", example: 1 })
  @IsInt()
  @IsPositive()
  id: number;
}
```

SquadPlayer DTO (im Squad DTO)

```
@SerializeOptions({ excludeExtraneousValues: true })
class SquadSquadPlayerDto {
  @ApiProperty({ description: "Spieler" })
  @ValidateNested()
  @IsInstance(PlayerId)
  @Type(() => PlayerId)
  player: PlayerId;

  @ApiProperty({ description: "Position vom Spieler" })
  position: Position;
}

@SerializeOptions({ excludeExtraneousValues: true })
class PlayerId {
  @ApiProperty({ description: "ID", example: 1 })
  @IsInt()
  @IsPositive()
  id: number;
}
```

Player Service

```
@Injectable()
class PlayerService {
  constructor(
    @InjectRepository(Player)
    private playerRepository: Repository<Player>,
  ) {}

  create(createPlayerDto: CreatePlayerDto): Promise<Player> {
    return this.playerRepository.save(createPlayerDto);
  }

  findAll(): Promise<Player[]> {
    return this.playerRepository.find();
  }

  findOne(id: number): Promise<Player | null> {
    return this.playerRepository.findOneBy({ id });
  }
  ...
```

```
  ...
  async update(
    id: number,
    updatePlayerDto: UpdatePlayerDto,
  ): Promise<Player | null> {
    const player = await this.findOne(id);
    if (!player) return null;
    const updatePlayer = this.playerRepository.merge(player, updatePlayerDto);
    return await this.playerRepository.save(updatePlayer);
  }

  async remove(id: number): Promise<boolean> {
    const exists = await this.playerRepository.existsBy({ id });
    if (exists) {
      await this.playerRepository.delete(id);
    }
    return exists;
  }
}
```

Trainer Service

```
@Injectable()
class TrainerService {
  constructor(
    @InjectRepository(Trainer)
    private trainerRepository: Repository<Trainer>,
  ) {}

  create(createTrainerDto: CreateTrainerDto): Promise<Trainer> {
    return this.trainerRepository.save(createTrainerDto);
  }

  findAll(): Promise<Trainer[]> {
    return this.trainerRepository.find();
  }

  findOne(id: number): Promise<Trainer | null> {
    return this.trainerRepository.findOneBy({ id });
  }
  ...
```

```
  ...
  async update(
    id: number,
    updateTrainerDto: UpdateTrainerDto,
  ): Promise<Trainer | null> {
    const trainer = await this.findOne(id);
    if (!trainer) return null;
    const updateTrainer = this.trainerRepository.merge(trainer, updateTrainerDto);
    return await this.trainerRepository.save(updateTrainer);
  }

  async remove(id: number): Promise<boolean> {
    const exists = await this.trainerRepository.existsBy({ id });
    if (exists) {
      await this.trainerRepository.delete(id);
    }
    return exists;
  }
}
```

Squad Service

```
@Injectable()
class SquadService {
  constructor(
    @InjectRepository(Squad)
    private squadRepository: Repository<Squad>,
  ) {}

  async create(createSquadDto: CreateSquadDto): Promise<Squad> {
    return Squad.new(await this.squadRepository.save(createSquadDto));
  }

  findAll(): Promise<Squad[]> {
    return this.squadRepository.find();
  }

  findOne(id: number): Promise<Squad | null> {
    return this.squadRepository.findOneBy({ id });
  }
  ...
```

```
  ...
  async update(
    id: number,
    updateSquadDto: UpdateSquadDto,
  ): Promise<Squad | null> {
    const squad = await this.findOne(id);
    if (!squad) return null;
    const updateSquad: DeepPartial<Squad> = { ...squad, ...updateSquadDto };
    return Squad.new(await this.squadRepository.save(updateSquad));
  }

  async remove(id: number): Promise<boolean> {
    const exists = await this.squadRepository.existsBy({ id });
    if (exists) {
      await this.squadRepository.delete(id);
    }
    return exists;
  }
}
```

Anforderungen

- Player:

Method	Pfad	Privat?	Beschreibung
`GET`	`/api/player/{id}`	protected	Auslesen einzelner und aller Spieler
`POST`	`/api/player`	protected	Anlegen einzelner Spieler*innen
`PUT`	`/api/player/{id}`	protected	Änderung einer bestehend Spieler*in
`DELETE`	`/api/player/{id}`	protected	Löschen einer Spieler*in

- Trainer:

Method	Pfad	Privat?	Beschreibung
`GET`	`/api/trainer/{id}`	protected	Auslesen einzelner und aller Trainer*innen
`POST`	`/api/trainer`	protected	Anlegen einzelner Trainer*innen
`PUT`	`/api/trainer/{id}`	protected	Änderung einer bestehend Trainer*in
`DELETE`	`/api/trainer/{id}`	protected	Löschen einer Trainer*in

Anforderungen

- Squad:

Method	Pfad	Privat?	Beschreibung
`GET`	`/api/squad/{id}`	public	Einzelner Teamkader
`POST`	`/api/squad`	protected	Neuen Squad anlegen
`PUT`	`/api/squad/{id}`	protected	Squad bearbeiten
`DELETE`	`/api/squad/{id}`	protected	Squad löschen
`GET`	`/api/squad`	protected	Liste aller existierenden Teamkader

Protected Endpoints sind mit einem JSON Web Token geschützt und sind nur für Benutzer, die sich bei `/api/login` eingeloggt haben, zugänglich.

Player Controller

```
@ApiUnauthorizedResponse()
@ApiBearerAuth()
@UseGuards(JwtGuard)
@UseInterceptors(ClassSerializerInterceptor)
@Controller("/player")
class PlayerController {
  constructor(private readonly playerService: PlayerService) {}

  @ApiOperation({ summary: "Erstelle einen Spieler" })
  @ApiCreatedResponse({ type: Player, description: "Spieler erstellt" })
  @Post()
  create(@Body() createPlayerDto: CreatePlayerDto): Promise<Player> {
    return this.playerService.create(createPlayerDto);
  }

  @ApiOperation({ summary: "Lese alle Spieler aus" })
  @ApiOkResponse({ type: [Player], description: "Alle Spieler ausgelesen" })
  @Get()
  findAll(): Promise<Player[]> {
    return this.playerService.findAll();
  }
```

```
@ApiOperation({ summary: "Lese einen Spieler aus" })
@ApiOkResponse({ type: Player, description: "Spieler ausgelesen" })
@ApiNotFoundResponse({ description: "Spieler existiert nicht" })
@Get(":id")
async findOne(@Param("id") id: number): Promise<Player> {
  const player = await this.playerService.findOne(id);
  if (!player) throw new NotFoundException();
  return player;
}

@ApiOperation({ summary: "Bearbeite einen Spieler" })
@ApiOkResponse({ type: Player, description: "Spieler bearbeitet" })
@ApiNotFoundResponse({ description: "Spieler existiert nicht" })
@Patch(":id")
async update(
  @Param("id") id: number,
  @Body() updatePlayerDto: UpdatePlayerDto,
): Promise<Player> {
  const player = await this.playerService.update(id, updatePlayerDto);
  if (!player) throw new NotFoundException();
  return player;
}
```

```
  @ApiOperation({ summary: "Lösche einen Spieler" })
  @ApiOkResponse({ description: "Spieler gelöscht" })
  @ApiNotFoundResponse({ description: "Spieler existiert nicht" })
  @Delete(":id")
  async remove(@Param("id") id: number): Promise<void> {
    const exists = await this.playerService.remove(id);
    if (!exists) throw new NotFoundException();
  }
}
```

Trainer Controller

```
@ApiUnauthorizedResponse()
@ApiBearerAuth()
@UseGuards(JwtGuard)
@UseInterceptors(ClassSerializerInterceptor)
@Controller("/trainer")
class TrainerController {
  constructor(private readonly trainerService: TrainerService) {}

  @ApiOperation({ summary: "Erstelle einen Trainer" })
  @ApiCreatedResponse({ type: Trainer, description: "Trainer erstellt" })
  @Post()
  create(@Body() createTrainerDto: CreateTrainerDto): Promise<Trainer> {
    return this.trainerService.create(createTrainerDto);
  }

  @ApiOperation({ summary: "Lese alle Trainer aus" })
  @ApiOkResponse({ type: [Trainer], description: "Alle Trainer ausgelesen" })
  @Get()
  findAll(): Promise<Trainer[]> {
    return this.trainerService.findAll();
  }
```

```
@ApiOperation({ summary: "Lese einen Trainer aus" })
@ApiOkResponse({ type: Trainer, description: "Trainer ausgelesen" })
@ApiNotFoundResponse({ description: "Trainer existiert nicht" })
@Get(":id")
async findOne(@Param("id") id: number): Promise<Trainer> {
  const trainer = await this.trainerService.findOne(id);
  if (!trainer) throw new NotFoundException();
  return trainer;
}

@ApiOperation({ summary: "Bearbeite einen Trainer" })
@ApiOkResponse({ type: Trainer, description: "Trainer bearbeitet" })
@ApiNotFoundResponse({ description: "Trainer existiert nicht" })
@Patch(":id")
async update(
  @Param("id") id: number,
  @Body() updateTrainerDto: UpdateTrainerDto,
): Promise<Trainer> {
  const trainer = await this.trainerService.update(id, updateTrainerDto);
  if (!trainer) throw new NotFoundException();
  return trainer;
}
```

```
  @ApiOperation({ summary: "Lösche einen Trainer" })
  @ApiOkResponse({ description: "Trainer gelöscht" })
  @ApiNotFoundResponse({ description: "Trainer existiert nicht" })
  @Delete(":id")
  async remove(@Param("id") id: number): Promise<void> {
    const exists = await this.trainerService.remove(id);
    if (!exists) throw new NotFoundException();
  }
}
```

Squad Controller

```
@Controller("/squad")
@UseInterceptors(ClassSerializerInterceptor)
class SquadController {
  constructor(private readonly squadService: SquadService) {}

  @ApiOperation({ summary: "Erstelle einen Squad" })
  @ApiCreatedResponse({ type: Squad, description: "Squad erstellt" })
  @ApiUnauthorizedResponse() @ApiBearerAuth() @UseGuards(JwtGuard)
  @Post()
  create(@Body() createSquadDto: CreateSquadDto): Promise<Squad> {
    return this.squadService.create(createSquadDto);
  }

  @ApiOperation({ summary: "Lese alle Squad aus" })
  @ApiOkResponse({ type: [Squad], description: "Alle Squads ausgelesen" })
  @ApiUnauthorizedResponse() @ApiBearerAuth() @UseGuards(JwtGuard)
  @Get()
  findAll(): Promise<Squad[]> {
    return this.squadService.findAll();
  }
```

```
  @ApiOperation({ summary: "Lese einen Squad aus" })
  @ApiOkResponse({ type: Squad, description: "Squad ausgelesen" })
  @ApiNotFoundResponse({ description: "Squad existiert nicht" })
  @Get(":id")
  async findOne(@Param("id") id: number): Promise<Squad | null> {
    const squad = await this.squadService.findOne(id);
    if (!squad) throw new NotFoundException();
    return squad;
  }

  @ApiOperation({ summary: "Bearbeite einen Squad" })
  @ApiOkResponse({ type: Squad, description: "Squad bearbeitet" })
  @ApiNotFoundResponse({ description: "Squad existiert nicht" })
  @ApiUnauthorizedResponse() @ApiBearerAuth() @UseGuards(JwtGuard)
  @Patch(":id")
  async update(
    @Param("id") id: number,
    @Body() updateSquadDto: UpdateSquadDto,
  ): Promise<Squad | null> {
    const squad = await this.squadService.update(id, updateSquadDto);
    if (!squad) throw new NotFoundException();
    return squad;
  }
```

```
  @ApiOperation({ summary: "Lösche einen Squad" })
  @ApiOkResponse({ description: "Squad gelöscht" })
  @ApiNotFoundResponse({ description: "Squad existiert nicht" })
  @ApiUnauthorizedResponse()
  @ApiBearerAuth()
  @UseGuards(JwtGuard)
  @Delete(":id")
  async remove(@Param("id") id: number): Promise<void> {
    const exists = await this.squadService.remove(id);
    if (!exists) throw new NotFoundException();
  }
}
```

Auth Service

```
@Injectable()
class AuthService {
  constructor(
    private readonly jwtService: JwtService,
    private readonly configService: ConfigService,
  ) {}
  validateUser(username: string, password: string): { username: string } | null {
    if (
      this.configService.get("jwt.username") === username &&
      this.configService.get("jwt.password") === password
    ) return { username };
    return null;
  }
  login(user: { username: string }): string {
    return this.jwtService.sign(user);
  }
}
```

Auth Controller

```
@Controller("/auth")
class AuthController {
  constructor(private readonly authService: AuthService) {}

  @ApiCreatedResponse({ type: JwtDto, description: "Login erfolgreich" })
  @ApiUnauthorizedResponse({ description: "Nicht berechtigt" })
  @Post("/login")
  login(@Body() body: LoginDto): JwtDto {
    const user = this.authService.validateUser(body.username, body.password);
    if (!user) {
      throw new UnauthorizedException("Invalid credentials");
    }
    return new JwtDto(this.authService.login(user));
  }
}
```

Testing (declarative tests)

```
const trainerTests: MockTest<Trainer> = {
  name: "trainer",
  entityClass: Trainer,
  entityModules: [TrainerModule],
  entities: [ { id: 1, firstName: "Yahya", lastName: "Sinwar", gender: Gender.Other },
    { id: 2, firstName: "Mohammed", lastName: "Deif", gender: Gender.Male },
    { id: 3, firstName: "Leila", lastName: "Khaled", gender: Gender.Female },
    { id: 4, firstName: "Hassan", lastName: "Nasrallah", gender: Gender.Male },
    { id: 5, firstName: "George", lastName: "Habash", gender: Gender.Male }, ],
  partialEntities: [ { firstName: "Yahya", lastName: "Sinwar" },
    { firstName: "Yahya", gender: Gender.Male },
    { lastName: "Sinwar", gender: Gender.Male }, ],
  invalidForCreateEntities: [
    { entity: { firstName: "Test" }, err: "missing last name and gender" },
    { entity: {}, err: "missing all properties" }, ],
  invalidEntities: [
    { entity: { firstName: "", lastName: "Test", gender: Gender.Male }, err: "empty first name" },
    { entity: { firstName: "Test", lastName: "", gender: Gender.Female }, err: "empty last name" },
    { entity: { firstName: "Test", lastName: "Player", gender: "INVALID" }, err: "invalid gender" },
    { entity: { firstName: "Test", lastName: "Player", gender: "" }, err: "invalid gender" } ],
};
```

Testing setup

```
const tests: MockTest[] = [playerTests, trainerTests, squadTests];
for (const test of tests) {
  describe(test.name, () => {
    let app: INestApplication<App>;
    const url = `/${test.name}`; // z.B.: "/player", "/trainer", "/squad"
    const mockRepository = newMockRepository();
    ...
    beforeAll(async () => {
      const moduleFixture0 = Test.createTestingModule({ imports: test.entityModules })
        .overrideProvider(getRepositoryToken(test.entityClass)).useValue(mockRepository);
      if (test.entityClassDepenedencies) {
        for (const entityClassDepenedency of test.entityClassDepenedencies) {
          moduleFixture0
            .overrideProvider(getRepositoryToken(entityClassDepenedency))
            .useValue(newMockRepository());
        }
      }
      const moduleFixture: TestingModule = await moduleFixture0
        .overrideGuard(JwtGuard).useValue(mockJwtGuard).compile();
      app = moduleFixture.createNestApplication();
      ...
```

Test (POST)

```
describe(`POST ${url}`, () => { // z.B.: "POST /player"
  for (const entity of test.entities) {
    it(`should create a ${test.name}`, () => { // z.B.: "should create a player"
      return request(app.getHttpServer())
        .post(url)
        .send(entity)
        .expect(201)
        .then((response) => expectObjectEq(entity, response.body));
    });
  }
  const invalidEntities = [...test.invalidForCreateEntities, ...test.invalidEntities];
  for (const { entity, err } of invalidEntities) {
    // z.B: "should fail to create a player because of missing last name and gender"
    it(`should fail to create a ${test.name} because of ${err}`, () => {
      return request(app.getHttpServer())
        .post(url)
        .send(entity)
        .expect(400);
    });
  }
})
```

Test (DELETE)

```
describe(`DELETE ${urlWithId(url, ":id")}`, () => {
  // z.B.: "should delete a trainer"
  it(`should delete a ${test.name}`, () => {
    mockRepository.existsBy.mockReturnValueOnce(true);
    return request(app.getHttpServer())
      .delete(urlWithId(url, 1))
      .expect(200);
  });

  // z.B.: "should fail to delete a trainer with 404 because it does not exist"
  it(`should fail to delete a ${test.name} with 404 because it does not exist`, () => {
    mockRepository.existsBy.mockReturnValueOnce(false);
    return request(app.getHttpServer())
      .delete(urlWithId(url, 1))
      .expect(404);
  });
});
```

Test (PATCH)

```
describe(`PATCH ${url}/:id`, () => {
  const entityToUpdate = test.entities[0];
  it(`should update a ${test.name}`, () => {
    mockRepository.findOneBy.mockReturnValueOnce(entityToUpdate);
    return request(app.getHttpServer())
      .patch(`${url}/${entityToUpdate.id}`)
      .send(entityToUpdate).expect(200)
      .then((response) => expectObjectEq(entityToUpdate, response.body));
  });
  for (const partialEntity of test.partialEntities) {
    it(`should update select fields of a ${test.name}`, () => {
      mockRepository.findOneBy.mockReturnValueOnce(entityToUpdate);
      return request(app.getHttpServer())
        .patch(`${url}/${entityToUpdate.id}`)
        .send(partialEntity).expect(200)
        .then((response) => {
          expectObjectEq({...entityToUpdate, ...partialEntity}, response.body);
        });
    });
  }
}
```